The University of Azad Jammu and Kashmir

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Roll NO: 2024-SE-14

Semester:4th

Course Title: Database System

Course Code: CS-22034

Submitted To: Engr. Awais Rathore

Department of Software Engineering

LAB 05

1. Introduction

This lab was performed to understand SQL filtering techniques using MySQL in XAMPP. Different SQL operators and clauses such as WHERE, AND, OR, BETWEEN, IN, LIKE, IS NULL, ORDER BY, and LIMIT were used to retrieve specific employee records from the Employee table.

2. Objective

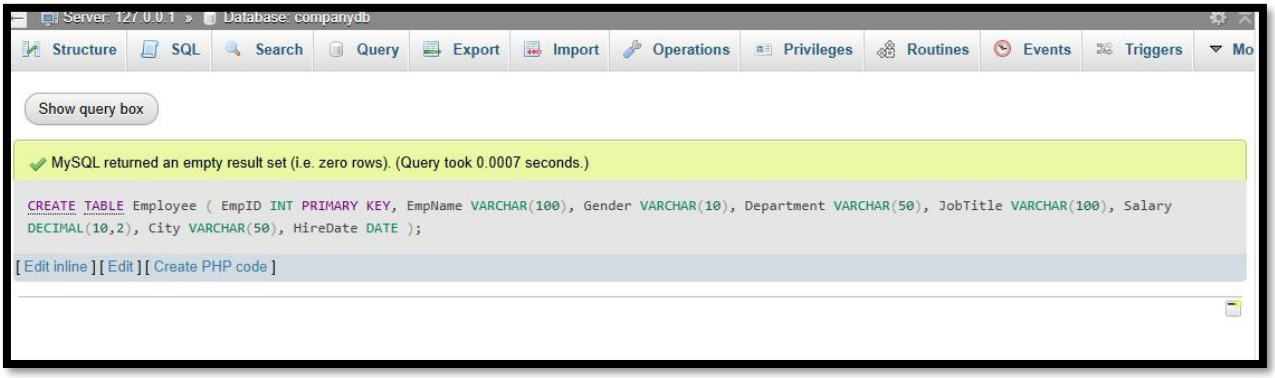
The objective of this lab was to practice SQL filtering and sorting operations on a database table using MySQL queries.

3. Database Creation

First, the XAMPP server was started and phpMyAdmin was opened through localhost. A database named **"companydb"** was created successfully.

4. Employee Table Creation

An Employee table was created in the companydb database using **SQL CREATE TABLE** statement.



Data Insertion

Sample employee records were inserted into the Employee table using INSERT INTO statement.

Employee records inserted successfully.

5. Part A Queries

Task A1

- This query displays employee ID, employee name, and salary of employees earning more than 90,000.

```
SELECT EmpID, EmpName, Salary
FROM Employee WHERE Salary > 90000;
```

Employees earning more than 90,000.

Showing rows 0 - 3 (4 total, Query took 0.0005 seconds.)

```
-- Task A1 SELECT EmpID, EmpName, Salary FROM Employee WHERE Salary > 90000;
```

[Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

	EmpID	EmpName	Salary
Edit Copy Delete	1	Ali Khan	95000.00
Edit Copy Delete	5	Usman Tariq	120000.00
Edit Copy Delete	8	Minal Fatima	99000.00
Edit Copy Delete	10	Maryam Imran	105000.00

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Output of Task A1 query.

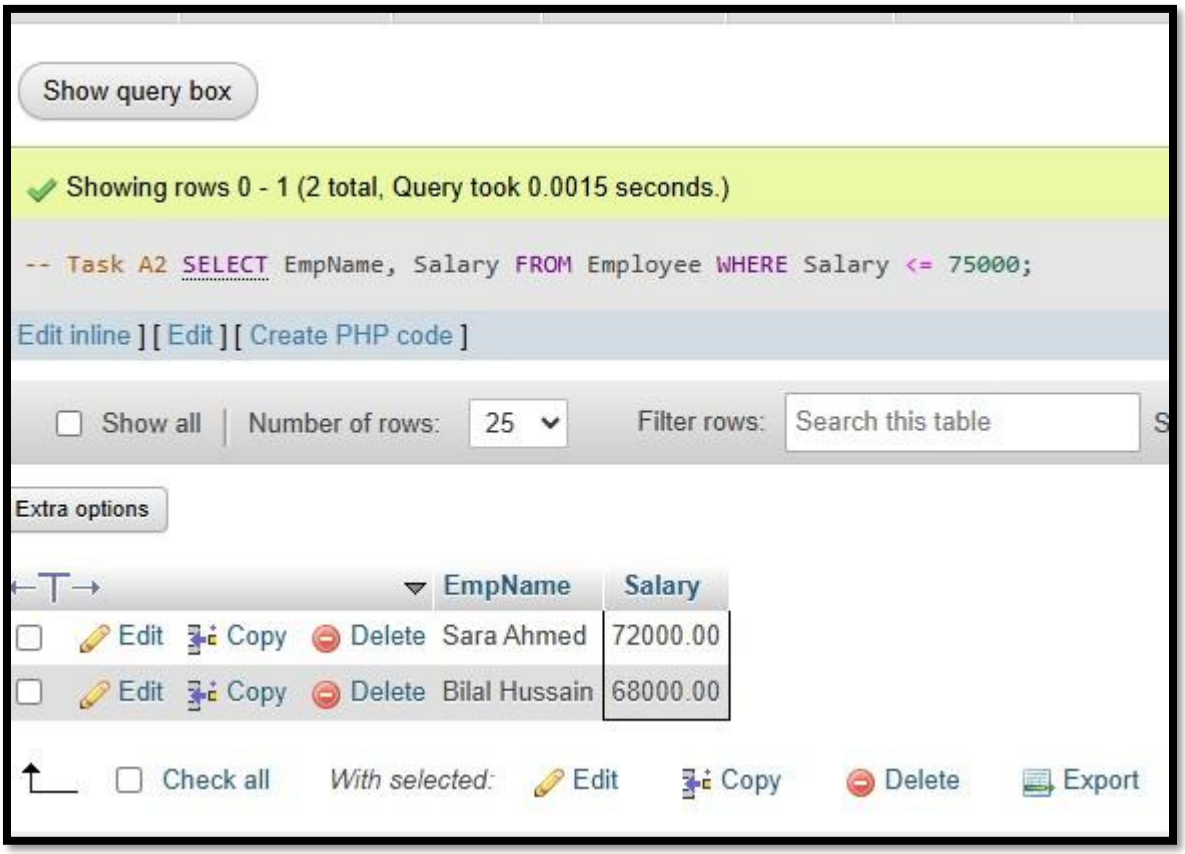
Task A2

- This query shows employees whose salary is less than or equal to75,000.

SELECT EmpName, Salary

FROM Employee

WHERE Salary <= 75000;

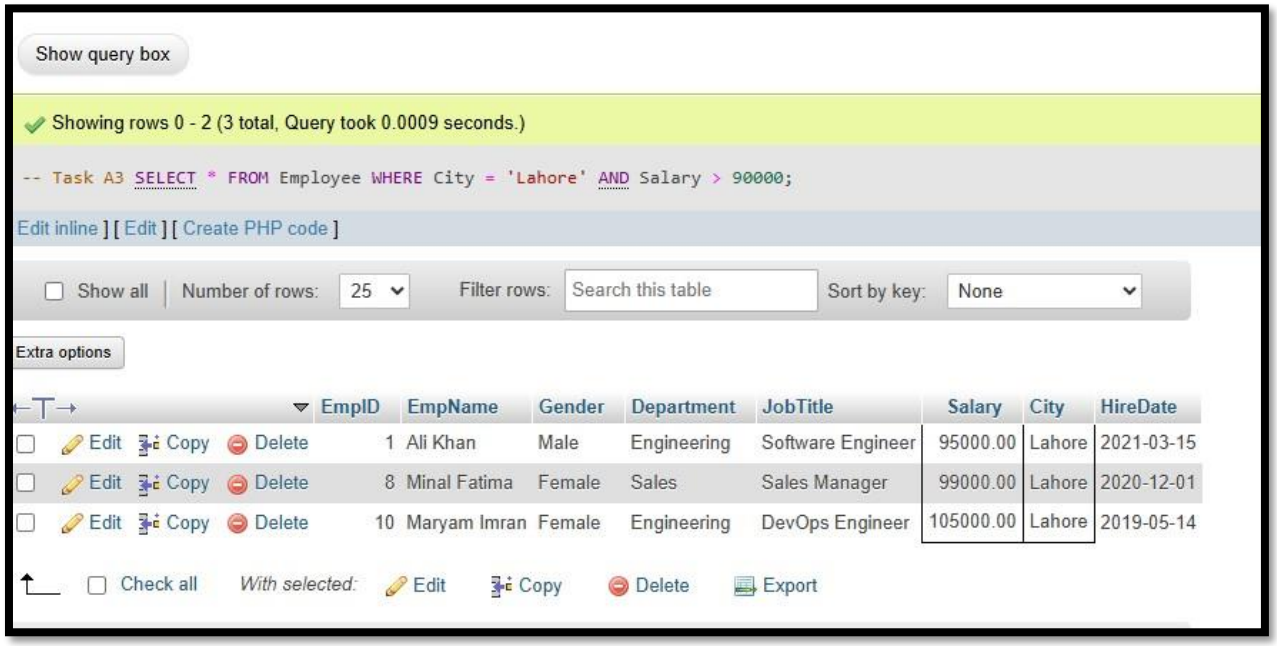


Output of Task A2 query showing employees with salary less than or equal to 75,000.

Task A3

- This query finds employees working in Lahore with salary greater than 90,000.

```
SELECT *
FROM Employee
WHERE City = 'Lahore'
AND Salary > 90000;
```



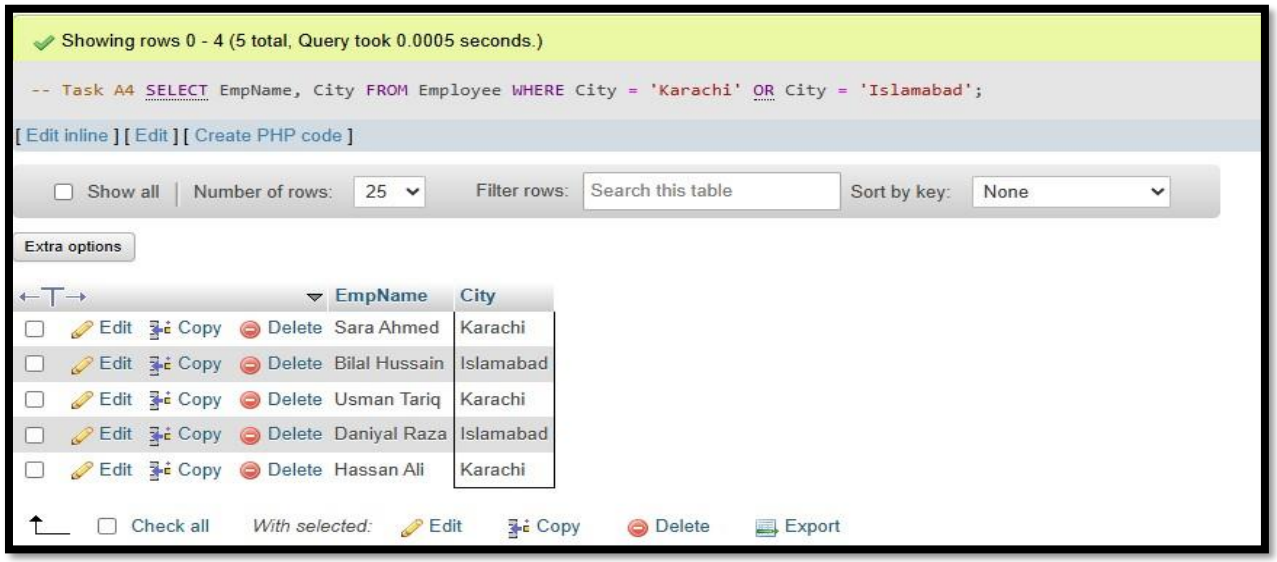
EmpID	EmpName	Gender	Department	JobTitle	Salary	City	HireDate
1	Ali Khan	Male	Engineering	Software Engineer	95000.00	Lahore	2021-03-15
8	Minal Fatima	Female	Sales	Sales Manager	99000.00	Lahore	2020-12-01
10	Maryam Imran	Female	Engineering	DevOps Engineer	105000.00	Lahore	2019-05-14

Output of Task A3 query showing employees from Lahore earning above 90,000.

Task A4

- This query lists employees belonging to Karachi or Islamabad.

```
SELECT EmpName, City FROM Employee
WHERE City = 'Karachi'
OR City = 'Islamabad';
```



Output of Task A4 query showing employees from Karachi or Islamabad.

Task A5

- This query displays female employees who are not part of the Engineering department.

```
SELECT * FROM Employee
        WHERE Gender = 'Female'
            AND Department <>
'Engineering';
```

Showing rows 0 - 2 (3 total, Query took 0.0006 seconds.)

```
-- Task A5 SELECT * FROM Employee WHERE Gender = 'Female' AND Department <> 'Engineering';
```

[Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	EmpID	EmpName	Gender	Department	JobTitle	Salary	City	HireDate
Edit Copy Delete	2	Sara Ahmed	Female	Marketing	Marketing Officer	72000.00	Karachi	2020-07-10
Edit Copy Delete	4	Maham Noor	Female	HR	HR Manager	88000.00	Lahore	2022-01-20
Edit Copy Delete	8	Minal Fatima	Female	Sales	Sales Manager	99000.00	Lahore	2020-12-01

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Output of Task A5 query showing female employees outside Engineering department.

Task A6

- This query finds male employees earning between 70,000 and 90,000.

```
SELECT *
FROM Employee
WHERE Gender = 'Male'
AND Salary >= 70000
AND Salary <= 90000;
```

Showing rows 0 - 1 (2 total, Query took 0.0005 seconds.)

```
-- Task A6 SELECT * FROM Employee WHERE Gender = 'Male' AND Salary >= 70000 AND Salary <= 90000;
```

[Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	EmpID	EmpName	Gender	Department	JobTitle	Salary	City	HireDate
Edit Copy Delete	7	Daniyal Raza	Male	Finance	Accountant	76000.00	Islamabad	2021-06-25
Edit Copy Delete	9	Hassan Ali	Male	Marketing	SEO Specialist	81000.00	Karachi	2022-08-30

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Output of Task A6 query showing male employees earning between 70,000 and 90,000.

Task A7

- This query lists employees who are Software Engineers or have salary greater than 100,000.

```
SELECT *
FROM Employee
WHERE JobTitle = 'Software Engineer'
OR Salary > 100000;
```

Showing rows 0 - 2 (3 total, Query took 0.0006 seconds.)

```
-- Task A7 SELECT * FROM Employee WHERE JobTitle = 'Software Engineer' OR Salary > 100000;
```

[Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	EmpID	EmpName	Gender	Department	JobTitle	Salary	City	HireDate
Edit Copy Delete	1	Ali Khan	Male	Engineering	Software Engineer	95000.00	Lahore	2021-03-15
Edit Copy Delete	5	Usman Tariq	Male	Engineering	Senior Software Engineer	120000.00	Karachi	2018-09-12
Edit Copy Delete	10	Maryam Imran	Female	Engineering	DevOps Engineer	105000.00	Lahore	2019-05-14

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Output of Task A7 query showing Software Engineers or employees earning above 100,000.

Task A8

- This query excludes employees from Marketing and Sales departments.

```
SELECT *
FROM Employee
WHERE Department <>
'Marketing'
AND Department <> 'Sales';
```

Showing rows 0 - 5 (6 total, Query took 0.0005 seconds.)

```
-- Task A8 SELECT * FROM Employee WHERE Department <> 'Marketing' AND Department <> 'Sales';
```

Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	EmpID	EmpName	Gender	Department	JobTitle	Salary	City	HireDate
Edit Copy Delete	1	Ali Khan	Male	Engineering	Software Engineer	95000.00	Lahore	2021-03-15
Edit Copy Delete	4	Maham Noor	Female	HR	HR Manager	88000.00	Lahore	2022-01-20
Edit Copy Delete	5	Usman Tariq	Male	Engineering	Senior Software Engineer	120000.00	Karachi	2018-09-12
Edit Copy Delete	6	Areeba Khan	Female	Engineering	QA Engineer	85000.00	NULL	2023-04-18
Edit Copy Delete	7	Daniyal Raza	Male	Finance	Accountant	76000.00	Islamabad	2021-06-25
Edit Copy Delete	10	Maryam Imran	Female	Engineering	DevOps Engineer	105000.00	Lahore	2019-05-14

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Output of Task A8 query excluding Marketing and Sales employees.

6. Part B Queries

Task B1

- This query displays employees whose salary lies between 75,000 and 100,000 in ascending order.

```
SELECT *
FROM Employee
WHERE Salary BETWEEN
75000 AND 100000
ORDER BY Salary ASC;
```

Showing rows 0 - 5 (6 total, Query took 0.0007 seconds.) [Salary: 76000.00... - 99000.00...]

```
-- Task B1 SELECT * FROM Employee WHERE Salary BETWEEN 75000 AND 100000 ORDER BY Salary ASC;
```

[Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	EmpID	EmpName	Gender	Department	JobTitle	Salary	City	HireDate
Edit Copy Delete	7	Daniyal Raza	Male	Finance	Accountant	76000.00	Islamabad	2021-06-25
Edit Copy Delete	9	Hassan Ali	Male	Marketing	SEO Specialist	81000.00	Karachi	2022-08-30
Edit Copy Delete	6	Areeba Khan	Female	Engineering	QA Engineer	85000.00	NULL	2023-04-18
Edit Copy Delete	4	Maham Noor	Female	HR	HR Manager	88000.00	Lahore	2022-01-20
Edit Copy Delete	1	Ali Khan	Male	Engineering	Software Engineer	95000.00	Lahore	2021-03-15
Edit Copy Delete	8	Minal Fatima	Female	Sales	Sales Manager	99000.00	Lahore	2020-12-01

Check all *With selected:* Edit Copy Delete Export

Output of Task B1 query showing employees with salary between 75,000 and 100,000.

Task B2

- This query lists employees hired between January 2020 and December 2022.

```
SELECT *
FROM Employee
WHERE HireDate
BETWEEN '2020-01-01' AND
'2022-12-31';
```

Showing rows 0 - 5 (6 total, Query took 0.0007 seconds.)

```
-- Task B2 SELECT * FROM Employee WHERE HireDate BETWEEN '2020-01-01' AND '2022-12-31';
```

[Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	EmpID	EmpName	Gender	Department	JobTitle	Salary	City	HireDate
Edit Copy Delete	1	Ali Khan	Male	Engineering	Software Engineer	95000.00	Lahore	2021-03-15
Edit Copy Delete	2	Sara Ahmed	Female	Marketing	Marketing Officer	72000.00	Karachi	2020-07-10
Edit Copy Delete	4	Maham Noor	Female	HR	HR Manager	88000.00	Lahore	2022-01-20
Edit Copy Delete	7	Daniyal Raza	Male	Finance	Accountant	76000.00	Islamabad	2021-06-25
Edit Copy Delete	8	Minal Fatima	Female	Sales	Sales Manager	99000.00	Lahore	2020-12-01
Edit Copy Delete	9	Hassan Ali	Male	Marketing	SEO Specialist	81000.00	Karachi	2022-08-30

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Output of Task B2 query showing employees hired between 2020 and 2022.

Task B3

- This query shows employees whose salary is outside the range of 80,000 to 100,000.

```
SELECT *
FROM Employee
WHERE Salary NOT
BETWEEN 80000 AND
100000;
```

Showing rows 0 - 4 (5 total, Query took 0.0005 seconds.)

```
-- Task B3 SELECT * FROM Employee WHERE Salary NOT BETWEEN 80000 AND 100000;
```

Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	EmpID	EmpName	Gender	Department	JobTitle	Salary	City	HireDate
Edit Copy Delete	2	Sara Ahmed	Female	Marketing	Marketing Officer	72000.00	Karachi	2020-07-10
Edit Copy Delete	3	Bilal Hussain	Male	Sales	Sales Executive	68000.00	Islamabad	2019-11-05
Edit Copy Delete	5	Usman Tariq	Male	Engineering	Senior Software Engineer	120000.00	Karachi	2018-09-12
Edit Copy Delete	7	Daniyal Raza	Male	Finance	Accountant	76000.00	Islamabad	2021-06-25
Edit Copy Delete	10	Maryam Imran	Female	Engineering	DevOps Engineer	105000.00	Lahore	2019-05-14

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Output of Task B3 query showing salaries outside 80,000 to 100,000.

Task B4

- This query lists employees from Lahore and Islamabad sorted by city and salary.

```
SELECT *
        FROM Employee
        WHERE City IN ('Lahore', 'Islamabad')
        ORDER BY City ASC, Salary DESC;
```

Show query box

Showing rows 0 - 5 (6 total, Query took 0.0006 seconds.) [City: ISLAMABAD... - LAHORE...] [Salary: 76000.00... - 88000.00...]

```
-- Task B4 SELECT * FROM Employee WHERE City IN ('Lahore', 'Islamabad') ORDER BY City ASC, Salary DESC;
```

[Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

←T→	EmpID	EmpName	Gender	Department	JobTitle	Salary 2	City 1	HireDate
Edit Copy Delete	7	Daniyal Raza	Male	Finance	Accountant	76000.00	Islamabad	2021-06-25
Edit Copy Delete	3	Bilal Hussain	Male	Sales	Sales Executive	68000.00	Islamabad	2019-11-05
Edit Copy Delete	10	Maryam Imran	Female	Engineering	DevOps Engineer	105000.00	Lahore	2019-05-14
Edit Copy Delete	8	Minal Fatima	Female	Sales	Sales Manager	99000.00	Lahore	2020-12-01
Edit Copy Delete	1	Ali Khan	Male	Engineering	Software Engineer	95000.00	Lahore	2021-03-15
Edit Copy Delete	4	Maham Noor	Female	HR	HR Manager	88000.00	Lahore	2022-01-20

Output of Task B4 query showing employees from Lahore and Islamabad.

Task B5

- This query excludes employees from Engineering, Sales, and HR departments.

```
SELECT *
FROM Employee
WHERE Department NOT IN
('Engineering', 'Sales', 'HR');
```

Show query box

Showing rows 0 - 2 (3 total, Query took 0.0006 seconds.)

```
-- Task B5 SELECT * FROM Employee WHERE Department NOT IN ('Engineering', 'Sales', 'HR');
```

[Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	EmpID	EmpName	Gender	Department	JobTitle	Salary	City	HireDate
Edit Copy Delete	2	Sara Ahmed	Female	Marketing	Marketing Officer	72000.00	Karachi	2020-07-10
Edit Copy Delete	7	Daniyal Raza	Male	Finance	Accountant	76000.00	Islamabad	2021-06-25
Edit Copy Delete	9	Hassan Ali	Male	Marketing	SEO Specialist	81000.00	Karachi	2022-08-30

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Output of Task B5 query excluding Engineering, Sales, and HR departments.

Task B6

- This query displays employee names starting with the letter M.

```
SELECT EmpName
FROM Employee WHERE EmpName LIKE 'M%';
```

Showing rows 0 - 2 (3 total, Query took 0.0005 seconds.)

```
-- Task B6 SELECT EmpName FROM Employee WHERE EmpName LIKE 'M%';
```

[Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	EmpName
Edit Copy Delete	Maham Noor
Edit Copy Delete	Minal Fatima
Edit Copy Delete	Maryam Imran

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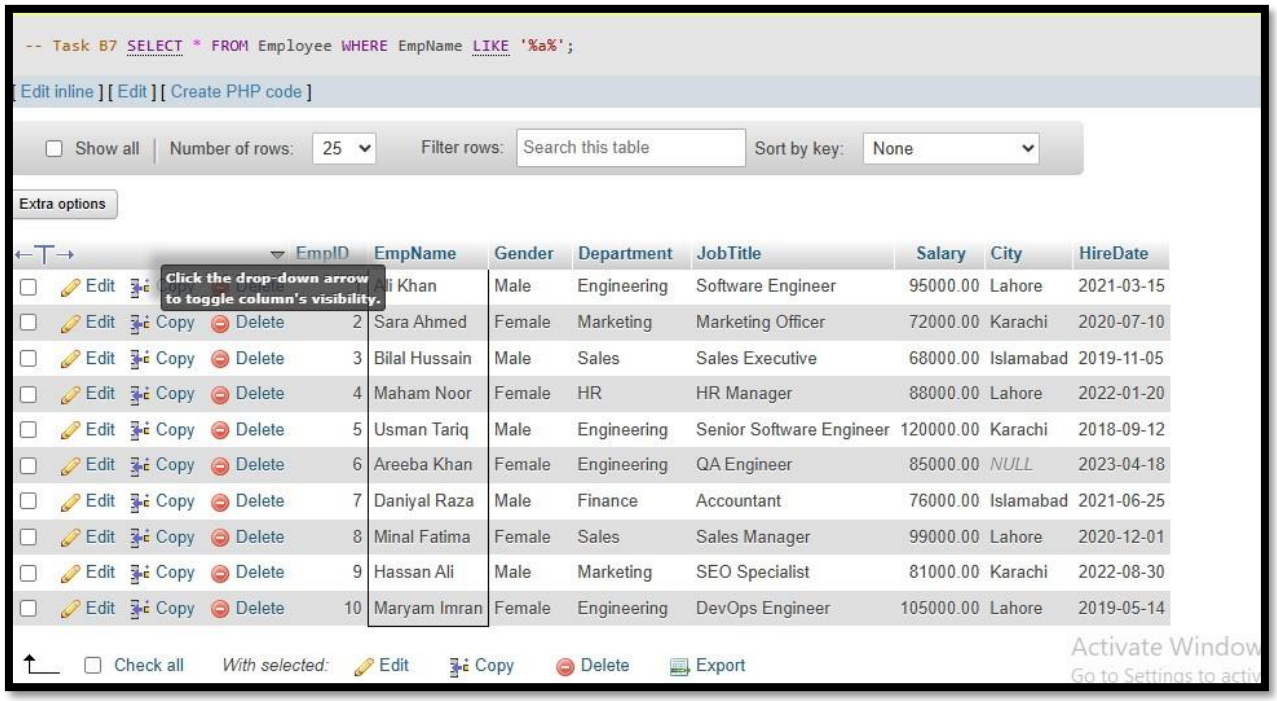
Output of Task B6 query showing names starting with M.

Task B7

- This query finds employee names containing the letter a.

```
SELECT *
FROM Employee
WHERE EmpName LIKE '%a%';
```



```
-- Task B7 SELECT * FROM Employee WHERE EmpName LIKE '%a%';
```

[Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

Click the drop-down arrow to toggle column's visibility.

	EmpID	EmpName	Gender	Department	JobTitle	Salary	City	HireDate
Edit		Ali Khan	Male	Engineering	Software Engineer	95000.00	Lahore	2021-03-15
Edit Copy Delete	2	Sara Ahmed	Female	Marketing	Marketing Officer	72000.00	Karachi	2020-07-10
Edit Copy Delete	3	Bilal Hussain	Male	Sales	Sales Executive	68000.00	Islamabad	2019-11-05
Edit Copy Delete	4	Maham Noor	Female	HR	HR Manager	88000.00	Lahore	2022-01-20
Edit Copy Delete	5	Usman Tariq	Male	Engineering	Senior Software Engineer	120000.00	Karachi	2018-09-12
Edit Copy Delete	6	Areeba Khan	Female	Engineering	QA Engineer	85000.00	NULL	2023-04-18
Edit Copy Delete	7	Daniyal Raza	Male	Finance	Accountant	76000.00	Islamabad	2021-06-25
Edit Copy Delete	8	Minal Fatima	Female	Sales	Sales Manager	99000.00	Lahore	2020-12-01
Edit Copy Delete	9	Hassan Ali	Male	Marketing	SEO Specialist	81000.00	Karachi	2022-08-30
Edit Copy Delete	10	Maryam Imran	Female	Engineering	DevOps Engineer	105000.00	Lahore	2019-05-14

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Activate Window
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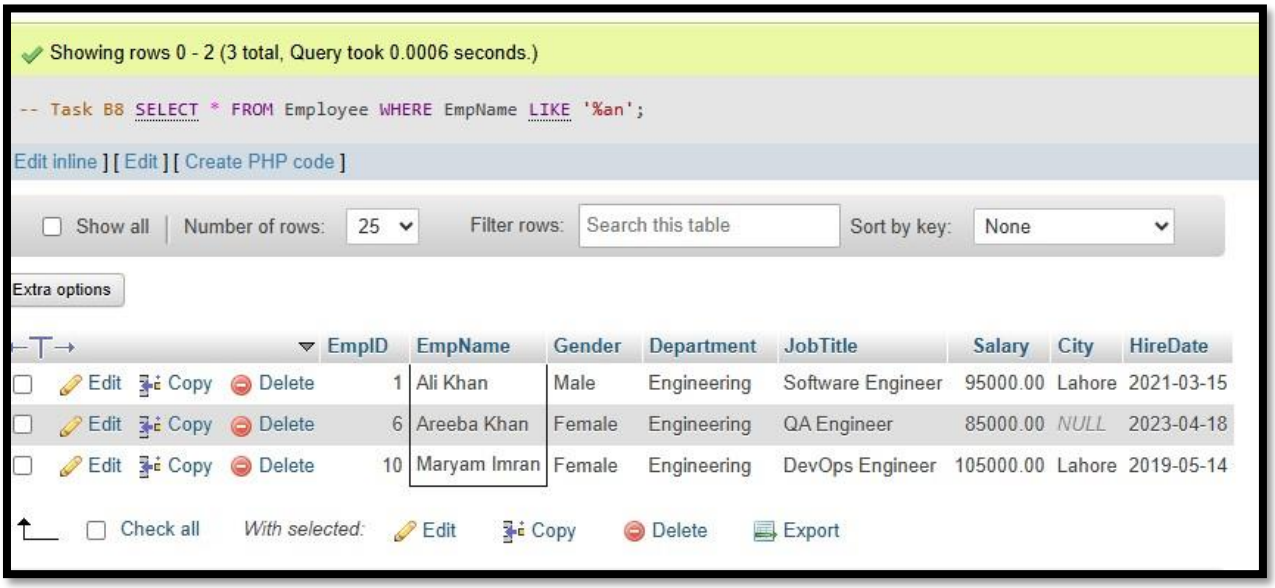
Output of Task B7 query showing names containing letter a.

Task B8

- This query lists employee names ending with the letters an.

```
SELECT *
FROM Employee
WHERE EmpName LIKE
'%an';
```

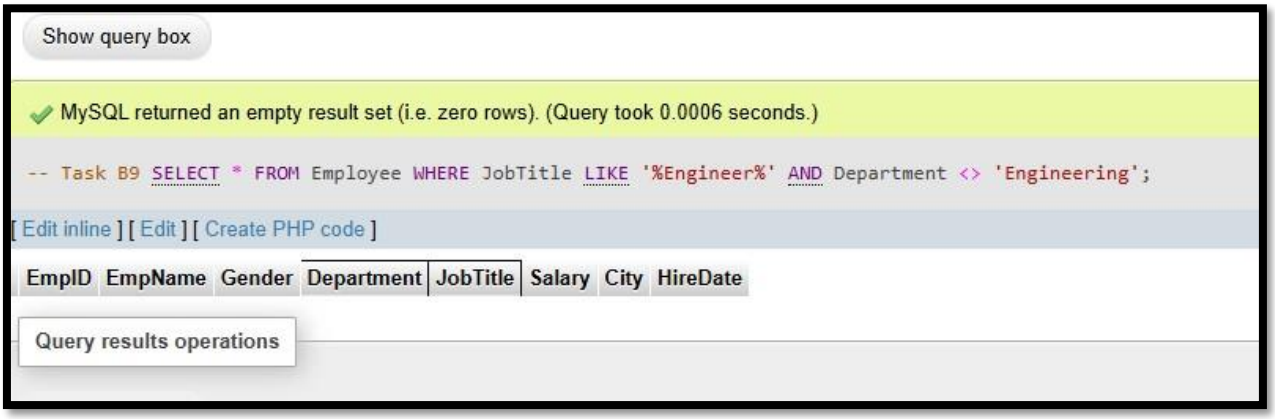


Output of Task B8 query showing names ending with an.

Task B9

- This query shows employees with Engineer in their job title but not working in Engineering department.

```
SELECT *
FROM Employee
WHERE JobTitle LIKE
'%Engineer%'
AND Department <>
'Engineering';
```



Output of Task B9 query showing engineers outside Engineering department.

Task B10

- This query lists employees who do not have a recorded city value.

```
SELECT EmpName
FROM Employee
WHERE City IS NULL;
```

Output of Task B10 query showing employees with NULL city values.

Task B11

- This query displays employees with available city values sorted alphabetically.

```
SELECT *
FROM Employee
WHERE City IS NOT
NULL
ORDER BY City ASC;
```

Showing rows 0 - 8 (9 total, Query took 0.0006 seconds.) [City: ISLAMABAD... - LAHORE...]

```
-- Task B11 SELECT * FROM Employee WHERE City IS NOT NULL ORDER BY City ASC;
```

[Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

←T→	EmpID	EmpName	Gender	Department	JobTitle	Salary	City 1	HireDate
Edit Copy Delete	3	Bilal Hussain	Male	Sales	Sales Executive	68000.00	Islamabad	2019-11-05
Edit Copy Delete	7	Daniyal Raza	Male	Finance	Accountant	76000.00	Islamabad	2021-06-25
Edit Copy Delete	2	Sara Ahmed	Female	Marketing	Marketing Officer	72000.00	Karachi	2020-07-10
Edit Copy Delete	5	Usman Tariq	Male	Engineering	Senior Software Engineer	120000.00	Karachi	2018-09-12
Edit Copy Delete	9	Hassan Ali	Male	Marketing	SEO Specialist	81000.00	Karachi	2022-08-30
Edit Copy Delete	1	Ali Khan	Male	Engineering	Software Engineer	95000.00	Lahore	2021-03-15
Edit Copy Delete	4	Maham Noor	Female	HR	HR Manager	88000.00	Lahore	2022-01-20
Edit Copy Delete	8	Minal Fatima	Female	Sales	Sales Manager	99000.00	Lahore	2020-12-01
Edit Copy Delete	10	Maryam Imran	Female	Engineering	DevOps Engineer	105000.00	Lahore	2019-05-14

Output of Task B11 query showing employees sorted by city.

Task B12

- This query shows the top three highest paid employees.

```
SELECT EmpName, Salary
FROM Employee ORDER BY
Salary DESC
        LIMIT 3;
```

Showing rows 0 - 2 (3 total, Query took 0.0006 seconds.) [Salary: 120000.00... - 99000.00...]

```
-- Task B12 SELECT EmpName, Salary FROM Employee ORDER BY Salary DESC LIMIT 3;
```

Edit inline] [Edit] [Create PHP code]

Extra options

	EmpName	Salary 1
Edit Copy Delete	Usman Tariq	120000.00
Edit Copy Delete	Maryam Imran	105000.00
Edit Copy Delete	Minal Fatima	99000.00

Check all *With selected:* Edit Copy Delete Export

Output of Task B12 query showing top three highest salaries.

Task B13

- This query lists the five most recently hired employees.

```
SELECT *
FROM Employee
ORDER BY HireDate DESC
LIMIT 5;
```

Showing rows 0 - 4 (5 total, Query took 0.0005 seconds.) [HireDate: 2023-04-18... - 2021-03-15...]

```
-- Task B13 SELECT * FROM Employee ORDER BY HireDate DESC LIMIT 5;
```

Edit inline] [Edit] [Create PHP code]

Extra options

	EmpID	EmpName	Gender	Department	JobTitle	Salary	City	HireDate 1
Edit Copy Delete	6	Areeba Khan	Female	Engineering	QA Engineer	85000.00	NULL	2023-04-18
Edit Copy Delete	9	Hassan Ali	Male	Marketing	SEO Specialist	81000.00	Karachi	2022-08-30
Edit Copy Delete	4	Maham Noor	Female	HR	HR Manager	88000.00	Lahore	2022-01-20
Edit Copy Delete	7	Daniyal Raza	Male	Finance	Accountant	76000.00	Islamabad	2021-06-25
Edit Copy Delete	1	Ali Khan	Male	Engineering	Software Engineer	95000.00	Lahore	2021-03-15

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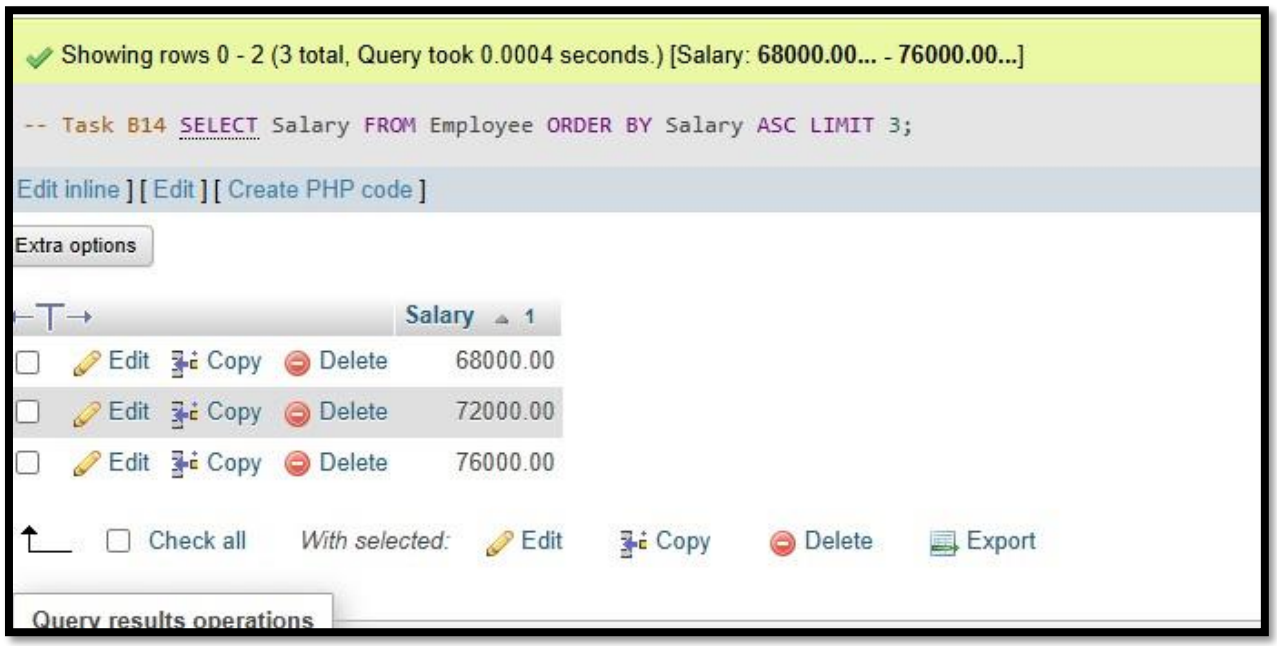
Output of Task B13 query showing five most recently hired employees.

Task B14

- This query displays the three lowest salaries in the company.

```
SELECT Salary
FROM Employee
ORDER BY Salary ASC
```



Output of Task B14 query showing the three lowest salaries.

Task B15

- This query sorts employees by department and hire date.

```
SELECT *
FROM Employee
ORDER BY Department
ASC, HireDate ASC;
```

```
-- Task B15 SELECT * FROM Employee ORDER BY Department ASC, HireDate ASC;
```

[Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	EmpID	EmpName	Gender	Department	JobTitle	Salary	City	HireDate
Edit Copy Delete	5	Usman Tariq	Male	Engineering	Senior Software Engineer	120000.00	Karachi	2018-09-12
Edit Copy Delete	10	Maryam Imran	Female	Engineering	DevOps Engineer	105000.00	Lahore	2019-05-14
Edit Copy Delete	1	Ali Khan	Male	Engineering	Software Engineer	95000.00	Lahore	2021-03-15
Edit Copy Delete	6	Areeba Khan	Female	Engineering	QA Engineer	85000.00	NULL	2023-04-18
Edit Copy Delete	7	Daniyal Raza	Male	Finance	Accountant	76000.00	Islamabad	2021-06-25
Edit Copy Delete	4	Maham Noor	Female	HR	HR Manager	88000.00	Lahore	2022-01-20
Edit Copy Delete	2	Sara Ahmed	Female	Marketing	Marketing Officer	72000.00	Karachi	2020-07-10
Edit Copy Delete	9	Hassan Ali	Male	Marketing	SEO Specialist	81000.00	Karachi	2022-08-30
Edit Copy Delete	3	Bilal Hussain	Male	Sales	Sales Executive	68000.00	Islamabad	2019-11-05
Edit Copy Delete	8	Minal Fatima	Female	Sales	Sales Manager	99000.00	Lahore	2020-12-01

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Output of Task B15 query showing employees sorted by department and hire date.

7. Conclusion

In this lab, different SQL filtering and sorting operations were performed successfully using MySQL in XAMPP. Various SQL clauses such as WHERE, AND, OR, BETWEEN, IN, LIKE, IS NULL, ORDER BY, and LIMIT were used to retrieve and manage employee records from the Employee table. This lab improved understanding of SQL query writing, data filtering techniques, pattern matching, and result sorting in.

*****THE END*****